

Predicate Logic Proof

Domain: All devices connected to a corporate network

Predicates:

$S(x)$: " x is sending suspicious traffic."

$I(x)$: " x must be investigated."

Premises:

$$1. \forall(x)(S(x) \rightarrow I(x))$$

$$2. \forall(x)(I(x) \rightarrow A(x))$$

$$3. S(\text{server-12})$$

Additional Predicate:

$A(x)$: " x generates an alert."

Task:

Prove: $\exists x A(x)$

Proof:

Step	Formula	Justification
1	$\forall x (S(x) \rightarrow I(x))$	Premise
2	$\forall x (I(x) \rightarrow A(x))$	Premise
3	$S(\text{server-12})$	Premise
4	$\forall x (S(x) \rightarrow A(x))$	Hypothetical Syllogism from (1) and (2)
5	$S(\text{Server-12}) \rightarrow A(\text{server-12})$	Universal instantiation from (3) and (4)
6	$A(\text{server-12})$	Modus Ponens from (3) and (5)
7	$\exists x A(x)$	Existential generalization from (6)

Predicate Logic Proof

Domain: All user accounts in a computer system.

Predicate:

- $A(x)$: " x has administrator privileges"
- $M(x)$: " x must be multi-factor authentication."

Premises:

$$1. \forall x (A(x) \rightarrow M(x))$$

$$2. A(\text{Admin-Alice})$$

Task:

Using the rules of predicate logic, prove the following conclusion:

$$\exists x M(x)$$

Proof:

step	Formula	Justification
1	$\forall x (A(x) \rightarrow M(x))$	Premise
2	$A(\text{Admin-Alice})$	Premise
3	$A(\text{Admin-Alice}) \rightarrow M(\text{Admin-Alice})$	Universal Instantiation from (1)
4	$M(\text{Admin-Alice})$	Modus Ponens from (2) and (3)
5	$\exists x M(x)$	Existential Generalization from (4)

